

Case 25: Spatially varying moisture quadrants

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE25_MQD. This case verifies spatial moisture lookup, moisture damping, and extinction behavior in otherwise uniform fuel. The verification question is whether the four regional spread rates meet their moisture-specific references, including negligible spread at the extinction condition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

Fuel moisture reduces reaction intensity through the model’s moisture-damping term; at or above the fuel’s moisture of extinction, sustainable spread approaches zero. Only the one-hour moisture raster varies, so regional velocity differences should follow that damping response.

For a dead-fuel moisture ratio $r = M_f/M_x$, the Rothermel damping polynomial used in the configured range is

$$\eta_M(r) = 1 - 2.59r + 5.11r^2 - 3.52r^3, \quad 0 \leq r \leq 1. \quad (1)$$

Reaction intensity contains this damping factor, $I_R = \Gamma'(w_{n,d}\eta_{M,d} + w_{n,l}\eta_{M,l})\eta_s h$, and the no-wind rate is $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$. At or above extinction, the dead-fuel contribution cannot sustain propagation.

3 Derivation of expected behavior

For Fuel Model 3 the configured moisture values 6, 12, 18, and 25 percent correspond to reference rates 1.511, 1.089, 0.801, and 0 m min⁻¹. The rates must decrease monotonically as moisture rises.

The dead one-hour moisture ratio is varied while all other generated fuel and moisture properties remain fixed. Applying $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ and $Q_{ig} = 250 + 1116M_f$ gives

M_{1h}	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
0.06	549.174	1.511830
0.12	479.466	1.089719
0.18	413.934	0.801066
0.25	$\simeq 3.9 \times 10^{-13}$	$\simeq 6.5 \times 10^{-16}$

so the report references are 1.511, 1.089, 0.801, and 0 m min⁻¹. Nonzero rows use $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the extinguished row uses the absolute residual $|R_{\max}|$, avoiding division by a zero reference.

4 Verification metrics and acceptance criteria

The maximum finite $\mathbf{vs}$ in each quadrant measures the local head response without averaging across moisture boundaries. Nonzero targets use 10 percent relative tolerance. The extinction quadrant uses an absolute 0.05 m min^{-1} limit because relative error is undefined at zero. All four components must pass.

5 Simulation configuration and predicted outcome

The grid uses uniform Fuel Model 3, flat terrain, zero wind, central ignition, and one-hour moisture quadrants of 6, 12, 18, and 25 percent. Predicted propagation slows monotonically and stops in the 25-percent quadrant.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1, t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

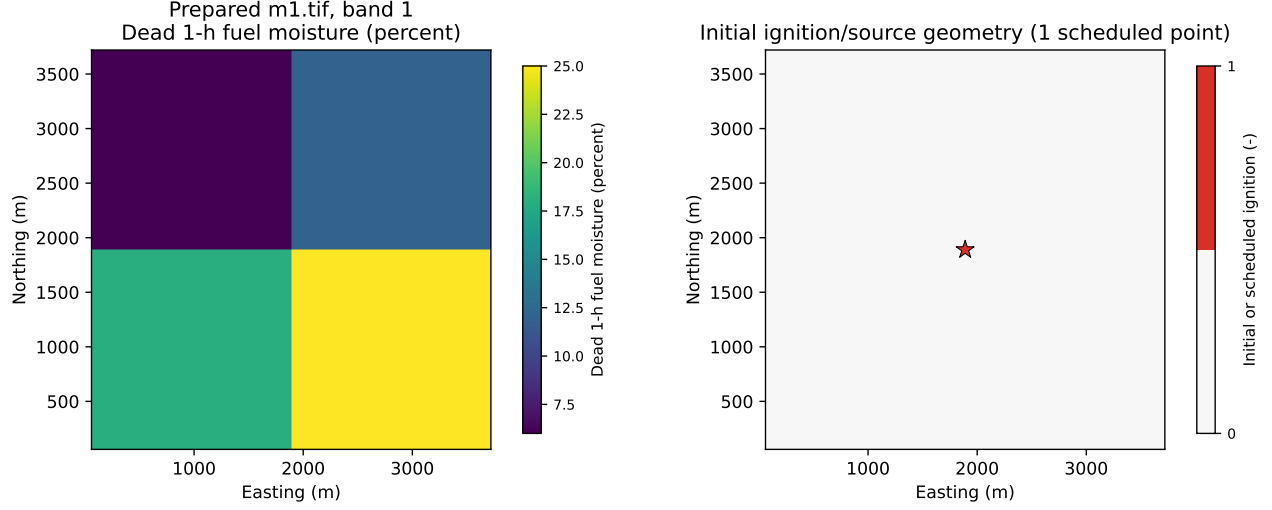


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

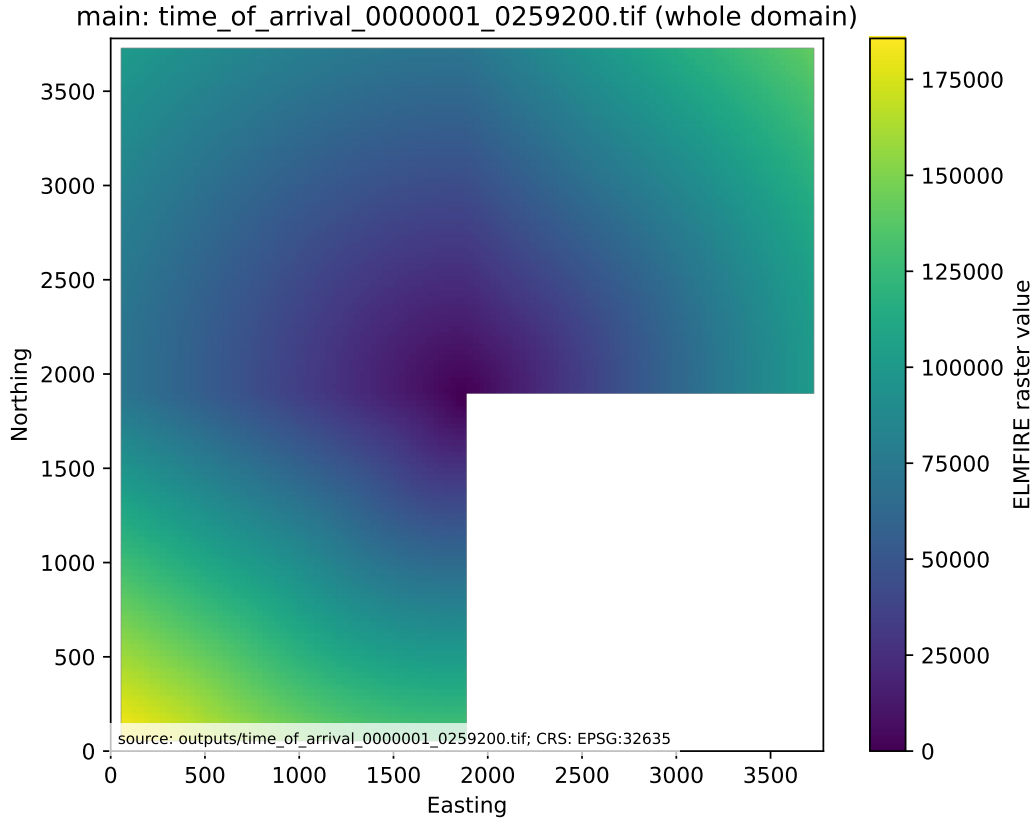


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / ROS @ 6%	1.511 m/min	1.51183 m/min	PASS
main / ROS @ 12%	1.089 m/min	1.08972 m/min	PASS
main / ROS @ 18%	0.801 m/min	0.801067 m/min	PASS
main / ROS @ 25%	0 m/min	0 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Nonmonotone behavior suggests moisture raster or damping errors; significant spread in the extinction quadrant is a direct failure. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

Only one fuel model and the one-hour dead-fuel moisture field are varied. This case provides implementation verification evidence only.